Math Club Seminar Series — Session #1

Principles of Mathematical Induction & Applications

Algebra • Calculus • Number Theory

Mathematical induction is a method for proving that a statement $P(n)$ is true for every natural number n, that is, that the infinitely many cases $P(0), P(1), P(2), P(3), \ldots$ all hold. This is done by first proving a simple case, then also showing that if we assume the claim is true for a given case, then the next case is also true. Informal metaphors help to explain this technique, such as falling dominoes or climbing a ladder:

> Mathematical induction proves that we can climb as high as we like on a ladder, by proving that we can climb onto the bottom rung (the **basis**) and that from each rung we can climb up to the next one (the **step**).
>
> —*Concrete Mathematics*, page 3 margins.



Mathematical induction can be informally illustrated by reference to the sequential effect of falling dominoes.[1][2]

A **proof by induction** consists of two cases. The first, the **base case**, proves the statement for $n = 0$ without assuming any knowledge of other cases. The second case, the **induction step**, proves that *if* the statement holds for any given case $n = k$, *then* it must also hold for the next case $n = k + 1$. These two steps establish that the statement holds for every natural number n. The base case does not necessarily begin with $n = 0$, but often with $n = 1$, and possibly with any fixed natural number $n = N$, establishing the truth of the statement for all natural numbers $n \geq N$.

Welcome to Math Club!

MISSION, SCOPE & APPROACH

What This Club Is About

- We go beyond standard school curricula to explore creative problem-solving techniques used in mathematics competitions worldwide.
- Algebra: Equational tricks, inequalities, and functional relations.
- Calculus & Analysis: Limits, infinite series, and rigorous convergence concepts.
- Number Theory: Modular arithmetic, prime structures, and integer representations.

How Our Meetings Work

- Core Concept Introduction: A rapid, high-yield conceptual foundation (15 mins).
- Competition Breakdown: Applying tools to real AMC, AIME, or Olympiad problems.
- Collaborative Solving: Group discussion and student-led whiteboard proofs.
- Weekly Challenge Puzzle: A top-tier competition puzzle to leave you thinking!

Part 1: Weak Mathematical Induction

CLASSICAL LADDER PROOFS

The Standard 2-Step Framework

- To prove P(n) is true for all integers $n \geq 1$:
- Base Case: Prove P(1) holds.
- Inductive Step: Assume P(k) is true (Inductive Hypothesis), then prove that P(k + 1) must follow.

Benchmark Example: Sum of Squares

- Prove that for all integers $n \geq 1$:
- $1^2 + 2^2 + 3^2 + \ldots + n^2 = n(n + 1)(2n + 1) / 6$
- Hint: Expand $(n + 1)^2$ after adding it to both sides of your inductive hypothesis sum.

AMC Practice Challenge

2023 AMC 12A — PROBLEM 12

Competition Application

Alternating Sum of Cubes Challenge

- Evaluate the exact expression from 2023 AMC 12A Q12:
- $2^3 - 1^3 + 4^3 - 3^3 + 6^3 - 5^3 + \ldots + 18^3 - 17^3$
- Your Tasks:

1. Group the terms into pairs of the form $(2k)^3 - (2k - 1)^3$.
2. Expand each difference using algebraic identities (difference of cubes or difference of squares).
3. Apply sum of squares and arithmetic series formulas to compute the final value!

- Hint: Notice that $(2k)^3 - (2k - 1)^3 = 12k^2 - 6k + 1$, which transforms the sum neatly.

Part 2: Strong Induction

WHEN ONE STEP BACK ISN'T ENOUGH

Strong Induction Proof Structure.

Start by saying what we want to prove: "Let $P(n)$ be the statement...."
Then establish two facts:

1. Base case: Prove that $P(0)$ is true. (Perhaps also prove other needed base cases.)
2. Inductive case: Assume $P(j)$ is true for all $j \leq k$. Prove that $P(k+1)$ is true.

Conclude, "Therefore, by strong induction, $P(n)$ is true for all $n > 0$."

Part 3: Advanced Induction Techniques

CAUCHY (FORWARD-BACKWARD) INDUCTION

Cauchy Induction

Cauch …overed …

Defi

For a … …al to … …e, and

that s … …ve m … …all

$n \geq 2$

Along … …or an … $f(n))$

for an … …o sh… …less

than … …nber, … ≥ 2.

The AM-GM Inequality Proof

- For non-negative real numbers a_1, a_2, ..., a_n:
- $(a_1 + a_2 + ... + a_n) / n \geq \sqrt[n]{(a_1 \; a_2 \; ... \; a_n)}$
- Augustin-Louis Cauchy proved this using a brilliant variant of induction!

How Cauchy Induction Works

- Base Step: Show P(2) holds.
- Forward Step (n → 2n): Prove that if P(k) holds, then P(2k) also holds. (Establishes truth for all powers of 2).
- Backward Step (n → n − 1): Prove that if P(k) holds, then P(k − 1) must also hold.
- Hint for Backward Step: Set a_n equal to the arithmetic mean of the previous terms to bridge the gap!

Find all positive integers n such that $n \times 2^n + 1$ is a square.

Solution

Let $a > 0$ be an integer such that $n \times 2^n + 1 = a^2$. It follows that $2^n n = (a-1)(a+1)$. The integers $a-1$ and $a+1$ have the same parity and their product is even, so both must be even. Moreover, since $a-1$ and $a+1$ are consecutive even integers, one of them is twice an odd number, so the other divisible by 2^{n-1}.

Thus $a+1 \geq 2^{n-1}$ which means $a-1 \leq 2n$. This implies that $2n+2 \geq a+1 \geq 2^{n-1}$.

We claim that this false for $n \geq 5$, so we only need to check $n = 1, 2, 3$ and 4.

When $n = 5$, we have $12 = 2n+2 < 2^{n-1} = 16$, so the claim holds for $n = 5$.

If we know that $2k+2 < 2^{k-1}$ for some positive k, we can consider:

$$2(k+1)+2 = (2k+2) \times \frac{k+2}{k+1} < 2^{k-1} \times \frac{k+2}{k+1} < 2^{k-1} \times 2 = 2^{(k+1)-1}$$

so if the claim holds for k, it holds for $k+1$.

Checking $n = 1, 2, 3$ and 4, we find that the values of $n \times 2^n + 1$ are $3, 9, 25$ and 65 respectively. Hence $n = 2$ and $n = 3$ work and $n = 1, 4$ do not.

Closing Grand Puzzle

BMO1 2015/16 — QUESTION 6

Olympiad Challenge

The 'Charming Numbers' Problem

- A positive integer is called charming if it is equal to 2, or if it is of the form $3^i 5^j$ where $i, j \geq 0$.
- Problem Statement: Prove that EVERY positive integer can be written as a sum of distinct charming integers.
- Discussion Hints for Group Work:
- • Check small base cases (1, 2, 3, 4, 5, 6).
- • Consider the property of charming number (introduce a notation for charming number)